

Algebraic Exponential Integrals: Transcendence and Linear Independence

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Abstract

Let $q \in \overline{\mathbb{Q}}[x]$ be nonconstant, let $\xi \in \overline{\mathbb{Q}}^\times$, and let $\Delta = \sum_\alpha d_\alpha[\alpha]$ be a finite degree-zero zero-chain with $d_\alpha, \alpha \in \overline{\mathbb{Q}}$. The entire function

$$I_{q,\Delta}(z) = \int_\Delta e^{zq(x)} dx$$

is defined by integration over any one-chain with boundary Δ . We prove the algebraic-value dichotomy

$$I_{q,\Delta}(\xi) \in \overline{\mathbb{Q}} \iff I_{q,\Delta}(z) \equiv 0.$$

Thus every nonzero algebraic-parameter value of an algebraic exponential integral is transcendental. More generally, all $\overline{\mathbb{Q}}$ -linear relations among finitely many such values, together with 1, are exactly the specializations of their functional relations. This yields linear independence, for example, for exponential integrals from a common endpoint to algebraic points whose q -values are pairwise distinct regular values.

For $\deg q \geq 2$, the proof uses an explicit inhomogeneous differential system for the moment functions $\int_\Delta x^j e^{zq(x)} dx$. A complete-reducibility theorem for its homogeneous part, combined with an elementary local-spectrum argument, implies that every stable rational relation space is obtained from a constant vector space. Beukers' refinement of the Siegel–Shidlovskii theorem then converts an algebraic value into an exponential-polynomial identity. A constant-coefficient annihilator and an edgewise Cauchy-transform argument force the original exponential period to vanish identically.

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1 Introduction and main results

For a polynomial phase $q \in \overline{\mathbb{Q}}[x]$ and algebraic endpoints A, B , consider

$$\int_A^B e^{\xi q(x)} dx, \quad \xi \in \overline{\mathbb{Q}}^\times. \quad (1.1)$$

When q is linear, the integral is an algebraic multiple of a difference of exponentials, and its transcendence follows from the Lindemann–Weierstrass theorem. For higher-degree phases, the integral is a special value of an E -function, but direct use of the Siegel–Shidlovskii theory is obstructed by possible functional relations among the auxiliary moment functions needed to form a first-order differential system. The purpose of this paper is to resolve that obstruction and to identify all algebraic linear relations among finite families of the values (1.1).

The arithmetic input is Beukers’ complete lifting theorem for relations among values of E -functions at nonsingular nonzero algebraic points [3]; see also the earlier linear-independence theorem of Nesterenko and Shidlovskii [8]. The general problem of exceptional algebraic values of E -functions was made effective by Adamczewski and Rivoal [1] and developed algorithmically through minimal differential equations by Bostan, Rivoal, and Salvy [4]. Delaygue recently proved a Lindemann–Weierstrass theorem for values of E -functions under a separation condition on the finite singularities of their associated G -functions [6]. The result here is complementary: the algebraic parameter is fixed, the boundary zero-chain varies, and the full relation space is identified without a singularity-separation hypothesis.

The structural input is a complete-reducibility theorem for the homogeneous moment system associated with the finite map $q : \mathbb{A}_x^1 \rightarrow \mathbb{A}_t^1$. We isolate that theorem from the arithmetic argument and formulate everything that follows in terms of matrices, stable rational subspaces, and rational projectors. Its standard proof is a short application of the decomposition theorem and Fourier transform for holonomic modules. The final analytic step is a Cauchy-transform rigidity argument in the spirit of the polynomial moment problem and of Cauchy-type integrals of algebraic functions [9, 10].

We first formulate the results for algebraic zero-chains. This both clarifies the exact linear-independence statement and records the unavoidable functional relations that can occur.

Definition 1.1 (Algebraic zero-chains and exponential periods). A finite algebraic zero-chain of degree zero is an expression

$$\Delta = \sum_{\alpha \in S} d_\alpha [\alpha], \quad S \subset \overline{\mathbb{Q}} \text{ finite}, \quad d_\alpha \in \overline{\mathbb{Q}}, \quad \sum_{\alpha \in S} d_\alpha = 0. \quad (1.2)$$

Choose a finite piecewise C^1 one-chain Γ in $\mathbb{C}$, with complex coefficients, such that $\partial\Gamma = \Delta$, and define

$$I_{q,\Delta}(z) = \int_{\Gamma} e^{zq(x)} dx. \quad (1.3)$$

The value is independent of Γ : the one-form $e^{zq(x)}dx$ is entire in x , and every closed one-chain in $\mathbb{C}$ has zero integral. Thus $I_{q,\Delta}$ depends only on q and Δ and is an entire function of z .

The notation in (1.3) may be read as integration “over the boundary data” Δ . For $\Delta = [B] - [A]$, it is simply

$$I_{q,[B]-[A]}(z) = \int_A^B e^{zq(x)} dx.$$

Theorem 1.2 (Algebraic-value rigidity). *Let $q \in \overline{\mathbb{Q}}[x]$ be nonconstant, let Δ be a finite algebraic zero-chain of degree zero, and let $\xi \in \overline{\mathbb{Q}}^\times$. Then*

$$I_{q,\Delta}(\xi) \in \overline{\mathbb{Q}} \iff I_{q,\Delta}(z) \equiv 0. \quad (1.4)$$

In the equivalent cases, $I_{q,\Delta}(\xi) = 0$. Equivalently, if $I_{q,\Delta}$ is not the zero function, then $I_{q,\Delta}(\xi)$ is transcendental for every $\xi \in \overline{\mathbb{Q}}^\times$.

Corollary 1.3 (Algebraic exponential-integral theorem). *Let $q \in \overline{\mathbb{Q}}[x]$ be nonconstant, let $A, B \in \overline{\mathbb{Q}}$ with $A \neq B$, and let $\xi \in \overline{\mathbb{Q}}^\times$. Then*

$$\int_A^B e^{\xi q(x)} dx$$

is transcendental. In particular, it is nonzero.

Indeed, $I_{q,[B]-[A]}(0) = B - A \neq 0$, so the corresponding entire function is not identically zero.

The next theorem is the exact linear-independence consequence of Theorem 1.2.

Theorem 1.4 (Transfer of linear relations). *Let $q \in \overline{\mathbb{Q}}[x]$ be nonconstant, let $\Delta_1, \dots, \Delta_r$ be finite algebraic zero-chains of degree zero, put $I_j(z) = I_{q,\Delta_j}(z)$, and let $\xi \in \overline{\mathbb{Q}}^\times$. For $a_0, a_1, \dots, a_r \in \overline{\mathbb{Q}}$, one has*

$$a_0 + \sum_{j=1}^r a_j I_j(\xi) = 0 \iff a_0 = 0 \text{ and } \sum_{j=1}^r a_j I_j(z) \equiv 0. \quad (1.5)$$

Consequently,

$$\dim_{\overline{\mathbb{Q}}} \text{span}_{\overline{\mathbb{Q}}} \{1, I_1(\xi), \dots, I_r(\xi)\} = 1 + \dim_{\overline{\mathbb{Q}}} \text{span}_{\overline{\mathbb{Q}}} \{I_1, \dots, I_r\}, \quad (1.6)$$

where the span on the right is taken in the vector space of entire functions.

Functional relations genuinely occur and therefore cannot be omitted from the statement. For example, if q is even, then

$$\int_a^b e^{zq(x)} dx + \int_{-a}^{-b} e^{zq(x)} dx = 0$$

identically in z . The following concrete criterion excludes this kind of collision.

Corollary 1.5 (Linear independence over separated regular values). *Let $a_0, a_1, \dots, a_r \in \overline{\mathbb{Q}}$ be such that $q(a_0), q(a_1), \dots, q(a_r)$ are pairwise distinct regular values of q . Then, for every $\xi \in \overline{\mathbb{Q}}^\times$, the numbers*

$$1, \quad \int_{a_0}^{a_1} e^{\xi q(x)} dx, \quad \dots, \quad \int_{a_0}^{a_r} e^{\xi q(x)} dx \quad (1.7)$$

are linearly independent over $\overline{\mathbb{Q}}$.

Here a regular value means a point of the target outside the finite set of critical values of q . We prove a more precise edge-density criterion for functional vanishing in Section 8; Corollary 1.5 follows from it in Section 9.

The proof of Theorem 1.2 has four stages. First, the moments

$$U_j(z) = \int_{\Delta} x^j e^{zq(x)} dx, \quad 0 \leq j \leq \deg q - 2,$$

form an explicit inhomogeneous first-order system and are E -functions. Second, complete reducibility and a rational-matrix rigidity theorem give a constant-splitting result for every stable relation space of that system. Third, Beukers' theorem and constant splitting show that an algebraic value of U_0 forces U_0 to be an exponential polynomial. Finally, a constant-coefficient annihilator produces a vanishing polynomial moment sequence, and an edgewise Cauchy-transform argument forces U_0 itself to vanish identically.

2 E -functions and value-relation lifting

We use the definition adopted by Beukers [3, Section 1].

Definition 2.1 (E -function). An entire function

$$F(z) = \sum_{m \geq 0} a_m \frac{z^m}{m!}$$

is an E -function if:

- (i) $a_m \in \overline{\mathbb{Q}}$ for every m ;
- (ii) the logarithmic height $h(a_0, \dots, a_m)$ is $O(m)$;
- (iii) F satisfies a nonzero linear differential equation with coefficients in $\overline{\mathbb{Q}}[z]$.

The decisive arithmetic input is the following specialization theorem.

Theorem 2.2 (Beukers). *Let $F_1, \dots, F_N$ be E -functions forming a vector solution of*

$$\mathbf{F}' = A(z)\mathbf{F}, \quad A(z) \in M_N(\overline{\mathbb{Q}}(z)),$$

and let $T(z) \in \overline{\mathbb{Q}}[z]$ be a common denominator of the entries of $A(z)$. Let $\xi \in \overline{\mathbb{Q}}$ satisfy $\xi T(\xi) \neq 0$. If $P \in \overline{\mathbb{Q}}[X_1, \dots, X_N]$ is homogeneous and

$$P(F_1(\xi), \dots, F_N(\xi)) = 0,$$

then there exists $Q \in \overline{\mathbb{Q}}[z, X_1, \dots, X_N]$, homogeneous in $X_1, \dots, X_N$ and of the same degree as P , such that

$$Q(z, F_1(z), \dots, F_N(z)) \equiv 0, \quad Q(\xi, X_1, \dots, X_N) = P(X_1, \dots, X_N).$$

In particular, a prescribed linear relation at $z = \xi$ lifts to a linear functional relation whose coefficient polynomials specialize coefficientwise to the prescribed relation.

This combines [3, Theorems 1.3 and 3.2]. Adjoining the constant E -function 1 converts affine relations into homogeneous ones.

We shall also use two standard independence statements.

Lemma 2.3 (Distinct exponential functions). *If $\lambda_1, \dots, \lambda_s \in \mathbb{C}$ are distinct, then $e^{\lambda_1 z}, \dots, e^{\lambda_s z}$ are linearly independent over $\mathbb{C}(z)$.*

Proof. Suppose a nontrivial relation with the fewest nonzero terms exists. Differentiate it and eliminate one term. Minimality forces the logarithmic derivative of a nonzero rational function to be a nonzero constant. This is impossible, since the Laurent expansion at infinity of a logarithmic derivative has zero constant term. $\square$

Theorem 2.4 (Lindemann–Weierstrass, linear form). *If $\beta_1, \dots, \beta_s \in \overline{\mathbb{Q}}$ are distinct, then $e^{\beta_1}, \dots, e^{\beta_s}$ are linearly independent over $\overline{\mathbb{Q}}$.*

We refer to [2, Chapter 1] for the classical theorem.

3 The moment differential system

Throughout Sections 3 to 6, let $q \in \overline{\mathbb{Q}}[x]$ have degree $n \geq 2$. No normalization of q is needed. For $0 \leq j \leq n-2$, perform Euclidean division

$$q(x)x^j = h_j(x)q'(x) + r_j(x), \quad \deg r_j \leq n-2. \quad (3.1)$$

Let $\mathbf{C}$ and $\mathbf{D}$ be the matrices whose j th columns are the coefficient vectors of r_j and $-h'_j$ in the basis $1, x, \dots, x^{n-2}$.

Lemma 3.1 (The two division matrices). *The following assertions hold.*

- (i) $\mathbf{C}$ is multiplication by q on the Jacobian algebra $\overline{\mathbb{Q}}[x]/(q')$;
- (ii) $\mathbf{D}$ is triangular with diagonal

$$-\frac{1}{n}, -\frac{2}{n}, \dots, -\frac{n-1}{n}.$$

Proof. The remainder r_j is precisely the representative of qx^j modulo q' of degree at most $n-2$, which proves (i). If the leading coefficient of q is a_n , then comparison of leading terms in (3.1) gives

$$h_j(x) = \frac{x^{j+1}}{n} + \text{a polynomial of degree at most } j.$$

Consequently

$$-h'_j(x) = -\frac{j+1}{n}x^j + \text{terms of degree at most } j-1.$$

Thus $\mathbf{D}$ is triangular with the stated diagonal. $\square$

Fix a finite algebraic zero-chain

$$\Delta = \sum_{\alpha \in S} d_\alpha [\alpha] \quad (3.2)$$

of degree zero. Choose a one-chain Γ with $\partial\Gamma = \Delta$, and define

$$U_j(z) = \int_{\Gamma} x^j e^{zq(x)} dx, \quad \mathbf{U} = (U_0, \dots, U_{n-2})^\top. \quad (3.3)$$

Define

$$\mathbf{b}(x) = n(h_0(x), \dots, h_{n-2}(x))^\top. \quad (3.4)$$

Let

$$\Lambda_\Delta = \{q(\alpha) : \alpha \in S, d_\alpha \neq 0\} \quad (3.5)$$

with repetitions removed, and for $\lambda \in \Lambda_\Delta$ put

$$\mathbf{b}_\lambda = \sum_{\substack{\alpha \in S \\ q(\alpha) = \lambda}} d_\alpha \mathbf{b}(\alpha). \quad (3.6)$$

Proposition 3.2 (Moment differential system). *Put $C = C^\top$ and $D = D^\top$. Then*

$$\mathbf{U}' = \left(C + \frac{D}{z}\right) \mathbf{U} + \frac{1}{nz} \sum_{\lambda \in \Lambda_\Delta} \mathbf{b}_\lambda e^{\lambda z}. \quad (3.7)$$

Moreover,

$$\text{Spec}(D) = \left\{-\frac{1}{n}, -\frac{2}{n}, \dots, -\frac{n-1}{n}\right\}. \quad (3.8)$$

Proof. Using (3.1) and integration by parts along Γ , we obtain, for $z \neq 0$,

$$\begin{aligned} U'_j &= \int_\Gamma q(x) x^j e^{zq(x)} dx \\ &= \int_\Gamma \left(r_j(x) - \frac{1}{z} h'_j(x)\right) e^{zq(x)} dx + \frac{1}{z} \sum_{\alpha \in S} d_\alpha h_j(\alpha) e^{zq(\alpha)}. \end{aligned}$$

Writing these identities simultaneously gives (3.7). The spectral assertion follows from Lemma 3.1. $\square$

4 The moment functions are E -functions

We record a height estimate that will be used to verify the arithmetic growth condition in Definition 2.1.

Lemma 4.1 (Joint height from one common denominator). *Let K_0 be a number field, let $\alpha_0, \dots, \alpha_m \in K_0$, and let $\delta \in \mathbb{Z}_{\geq 1}$ satisfy $\delta \alpha_k \in \mathcal{O}_{K_0}$ for every k . Then*

$$h(\alpha_0, \dots, \alpha_m) \leq \log \delta + \log \max \left(1, \max_{0 \leq k \leq m} \max_{\sigma: K_0 \hookrightarrow \mathbb{C}} |\sigma(\alpha_k)|\right).$$

Here the left side is the absolute logarithmic Weil height of $(1 : \alpha_0 : \dots : \alpha_m)$.

Proof. With the standard absolute values and local degrees $n_v = [(K_0)_v : \mathbb{Q}_v]$,

$$h(\alpha_0, \dots, \alpha_m) = \frac{1}{[K_0 : \mathbb{Q}]} \sum_{v \in M_{K_0}} n_v \log \max(1, |\alpha_0|_v, \dots, |\alpha_m|_v).$$

Put $\beta_k = \delta \alpha_k \in \mathcal{O}_{K_0}$. Projective invariance gives

$$(1 : \alpha_0 : \dots : \alpha_m) = (\delta : \beta_0 : \dots : \beta_m).$$

At every finite place all displayed coordinates have absolute value at most 1. At an infinite place,

$$\max(|\delta|_v, |\beta_0|_v, \dots, |\beta_m|_v) \leq \delta \max \left(1, \max_{k, \sigma} |\sigma(\alpha_k)|\right).$$

Summing over the infinite places proves the claim. $\square$

Lemma 4.2 (Moment functions are E -functions). *For $0 \leq j \leq n-2$, the entire function U_j in (3.3) is an E -function. More precisely, if*

$$U_j(z) = \sum_{m \geq 0} u_{j,m} \frac{z^m}{m!}, \quad u_{j,m} = \int_{\Gamma} x^j q(x)^m dx, \quad (4.1)$$

then there is a constant κ_j such that

$$h(u_{j,0}, \dots, u_{j,m}) \leq \kappa_j(m+1) \quad (m \geq 0),$$

and U_j satisfies a nonzero homogeneous linear differential equation over $\overline{\mathbb{Q}}[z]$ of order at most $n-1+|\Lambda_{\Delta}|$.

Proof. Uniform convergence on compact subsets permits termwise integration of $e^{zq(x)}$ along Γ , proving (4.1) and the entirety of U_j .

Let

$$K_0 = \mathbb{Q}(\{d_{\alpha}, \alpha : \alpha \in S\}, \text{the coefficients of } q).$$

Write

$$q(x) = \sum_{i=0}^n c_i x^i, \quad q(x)^m = \sum_{k=0}^{nm} \gamma_{m,k} x^k.$$

Since a primitive of x^{j+k} is $x^{j+k+1}/(j+k+1)$, the boundary formula for integration gives the exact identity

$$u_{j,m} = \sum_{k=0}^{nm} \gamma_{m,k} \frac{\sum_{\alpha \in S} d_{\alpha} \alpha^{j+k+1}}{j+k+1}. \quad (4.2)$$

Thus $u_{j,m} \in K_0$.

We first bound all archimedean conjugates simultaneously. Put

$$M = \max_{\substack{\sigma: K_0 \hookrightarrow \mathbb{C} \\ \alpha \in S}} \max(1, |\sigma(\alpha)|), \quad \Theta = \max_{\sigma: K_0 \hookrightarrow \mathbb{C}} \max_{|x| \leq M} |q^{\sigma}(x)|,$$

and

$$D_0 = \max_{\sigma: K_0 \hookrightarrow \mathbb{C}} \sum_{\alpha \in S} |\sigma(d_{\alpha})|.$$

If $P_{j,m}^{\sigma}(x) = \int_0^x t^j q^{\sigma}(t)^m dt$ is taken along the straight segment, then applying σ to (4.2) gives

$$\sigma(u_{j,m}) = \sum_{\alpha \in S} \sigma(d_{\alpha}) P_{j,m}^{\sigma}(\sigma(\alpha)).$$

Every such segment lies in $|x| \leq M$, and hence

$$|\sigma(u_{j,m})| \leq D_0 M^{j+1} \Theta^m.$$

This is bounded by C_1^{m+j+2} for a constant $C_1 \geq 1$ independent of m and σ .

Choose $d \in \mathbb{Z}_{\geq 1}$ so that

$$dc_0, \dots, dc_n, \quad d\alpha, \quad dd_{\alpha} \in \mathcal{O}_{K_0} \quad (\alpha \in S).$$

For $m \geq 0$, set

$$\delta_m = d^{(n+1)m+j+2} \operatorname{lcm}(1, \dots, nm + j + 1). \quad (4.3)$$

Then $d^m \gamma_{m,k}$, dd_α , and $d^{j+k+1} \alpha^{j+k+1}$ are algebraic integers. Since $0 \leq k \leq nm$, each summand in (4.2), after multiplication by δ_m , is a sum of products of algebraic integers; the spare factor d^{nm-k} supplies the uniform exponent in (4.3). Therefore $\delta_m u_{j,m}$ is integral. Moreover, $\delta_\ell \mid \delta_m$ for $0 \leq \ell \leq m$, so δ_m is a common denominator for $u_{j,0}, \dots, u_{j,m}$.

By Lemma 4.1,

$$h(u_{j,0}, \dots, u_{j,m}) \leq \log \delta_m + (m + j + 2) \log C_1.$$

The elementary Chebyshev estimate $\log \operatorname{lcm}(1, \dots, N) \leq 2N$ gives $\log \delta_m = O(m)$, and hence the required joint-height bound.

It remains to establish a scalar differential equation. Let $\lambda_1, \dots, \lambda_s$ be the distinct elements of Λ_Δ . By Proposition 3.2, the $\overline{\mathbb{Q}}(z)$ -span of

$$U_0, \dots, U_{n-2}, e^{\lambda_1 z}, \dots, e^{\lambda_s z}$$

is stable under ∂_z and has dimension at most $n - 1 + s$. Therefore U_j and its first $n - 1 + s$ derivatives are linearly dependent over $\overline{\mathbb{Q}}(z)$. Clearing denominators yields a nonzero operator in $\overline{\mathbb{Q}}[z]\langle \partial_z \rangle$ of order at most $n - 1 + s$ annihilating U_j . All three defining conditions of an E -function now follow. $\square$

5 Complete reducibility and constant splitting

Put

$$A(z) = C + \frac{D}{z} \quad (5.1)$$

and define, on rational row vectors,

$$\mathcal{T}(\phi) = \phi' + \phi A(z). \quad (5.2)$$

A $\mathbb{C}(z)$ -subspace of $\mathbb{C}(z)^{1 \times (n-1)}$ will be called stable if it is preserved by $\mathcal{T}$. We call the system completely reducible if every stable subspace has a stable complement.

Proposition 5.1 (Complete reducibility). *The row system (5.2) is completely reducible.*

Proof. We give the standard geometric proof, but the remainder of the paper uses only the statement of the proposition. Let Σ be the finite set of critical values of q , put $V = \mathbb{A}_t^1 \setminus \Sigma$, and let $\mathcal{L}$ be the permutation local system of the unramified covering $q : q^{-1}(V) \rightarrow V$. The covering is connected, so its monodromy action is transitive. The trace map splits

$$\mathcal{L} \simeq \mathbb{C}_V \oplus \mathcal{L}^0,$$

and $\mathcal{L}^0$ is semisimple by Maschke's theorem, since the monodromy image is a finite subgroup of S_n .

The map $q : \mathbb{A}_x^1 \rightarrow \mathbb{A}_t^1$ is finite, proper, and small. Hence

$$Rq_* \mathbb{C}_{\mathbb{A}_x^1}[1] \simeq \operatorname{IC}(\mathcal{L});$$

see [5, Remark 2.13]. Under the Riemann–Hilbert correspondence and compatibility with proper direct image, this yields

$$q_+ \mathcal{O}_{\mathbb{A}_x^1} \simeq \mathcal{O}_{\mathbb{A}_t^1} \oplus \mathcal{N}^0, \quad (5.3)$$

where $\mathcal{N}^0$ is a direct sum of simple regular holonomic modules; see [7, Sections 3.4, 7.1–7.2, and 8.2.2].

Use a Fourier variable τ with

$$t \longmapsto -\partial_\tau, \quad \partial_t \longmapsto \tau. \quad (5.4)$$

This Weyl-algebra automorphism is exact and preserves simple summands. The Fourier transform of $\mathcal{O}_{\mathbb{A}^1_t}$ is supported at $\tau = 0$ and vanishes after localization to $\mathbb{C}(\tau)$. We also record the elementary localization point. If M is a simple Weyl module and $S = \mathbb{C}[\tau] \setminus \{0\}$, then the S -torsion of M is a Weyl submodule: from $p(\tau)m = 0$ one obtains $p(\tau)^2\partial_\tau m = 0$. Hence either $S^{-1}M = 0$, or M embeds in $S^{-1}M$. In the latter case, every nonzero submodule of $S^{-1}M$ meets M after clearing a denominator, and simplicity forces that submodule to be all of $S^{-1}M$. Thus the surviving generic Fourier transform of $\mathcal{N}^0$ is completely reducible.

For completeness, we identify it explicitly. The graph module of q is generated by a symbol δ_q satisfying

$$(t - q(x))\delta_q = 0, \quad (\partial_x + q'(x)\partial_t)\delta_q = 0.$$

After (5.4), its transform satisfies

$$(\partial_\tau + q(x))\widehat{\delta}_q = 0, \quad (\partial_x + \tau q'(x))\widehat{\delta}_q = 0.$$

On coefficient polynomials multiplying $\widehat{\delta}_q$, the two relevant operators are therefore $\partial_x - \tau q'(x)$ and $\partial_\tau - q(x)$. Projection along the x -line is computed by the relative complex

$$\mathbb{C}(\tau)[x] \xrightarrow{\partial_x - \tau q'} \mathbb{C}(\tau)[x].$$

Its kernel is zero, because for nonzero $p \in \mathbb{C}(\tau)[x]$ the term $-\tau q'p$ has strictly larger x -degree than p' . Hence the generic transform is

$$\mathbb{C}(\tau)[x] / (\partial_x - \tau q'(x))\mathbb{C}(\tau)[x], \quad \partial_\tau[p] = [\partial_\tau p - qp]. \quad (5.5)$$

Pulling back by $\tau = -z$ gives

$$\mathbb{C}(z)[x] / (\partial_x + zq'(x))\mathbb{C}(z)[x], \quad \partial_z[p] = [\partial_z p + qp]. \quad (5.6)$$

The classes of $1, x, \dots, x^{n-2}$ form a basis: the relation $[zq'p] = -[p']$ reduces every higher power, while the image of a nonzero polynomial has degree larger by $n-1$ and therefore contains no nonzero polynomial of degree at most $n-2$. In this basis, (3.1) gives

$$\partial_z[x^j] = [qx^j] = [r_j] - \frac{1}{z}[h'_j],$$

so the matrix is $\mathbf{C} + \mathbf{D}/z$. The moment matrix is its transpose, $A(z) = C + D/z$. Complete reducibility is preserved by transposition and by passing to the dual row system, proving the proposition. $\square$

The next result is elementary and is the only special property of the matrices C, D needed to turn complete reducibility into constant splitting. Let

$$\mathcal{V} = \mathbb{C}[x]_{<n-1}.$$

For $g \in \mathcal{V}$, divide

$$gq = h_g q' + r_g, \quad \deg r_g < n - 1, \quad (5.7)$$

and define

$$\mathbf{C}_* g = r_g, \quad \mathbf{D}_* g = -h'_g. \quad (5.8)$$

These operators are represented by $\mathbf{C}$ and $\mathbf{D}$ in the monomial basis. For a critical point τ , put

$$m_\tau = \text{ord}_\tau q', \quad \lambda_\tau = q(\tau).$$

Lemma 5.2 (Critical-point basis and local spectrum). *For every critical point τ and $1 \leq k \leq m_\tau$, define*

$$g_{\tau,k}(x) = \frac{q'(x)}{(x - \tau)^k}. \quad (5.9)$$

Then:

(i) *the polynomials $g_{\tau,k}$, over all τ and k , form a basis of $\mathcal{V}$;*

(ii) *$\mathbf{C}_*$ is semisimple and*

$$\mathbf{C}_* g_{\tau,k} = \lambda_\tau g_{\tau,k};$$

(iii) *put*

$$E_\tau = \text{span}_{\mathbb{C}}\{g_{\tau,1}, \dots, g_{\tau,m_\tau}\}.$$

If $E_\lambda = \ker(\mathbf{C}_ - \lambda I)$ and π_λ is the spectral projection, then*

$$E_\lambda = \bigoplus_{\tau: \lambda_\tau = \lambda} E_\tau, \quad \overline{\mathbf{D}}_\lambda = \pi_\lambda \mathbf{D}_*|_{E_\lambda}$$

preserves every summand E_τ . In the ordered basis $g_{\tau,1}, \dots, g_{\tau,m_\tau}$, its restriction to E_τ is triangular with diagonal

$$-\frac{m_\tau}{m_\tau + 1}, -\frac{m_\tau - 1}{m_\tau + 1}, \dots, -\frac{1}{m_\tau + 1}. \quad (5.10)$$

Proof. Factor

$$q'(x) = c \prod_{\tau} (x - \tau)^{m_\tau}.$$

Reduction modulo q' identifies $\mathcal{V}$ with $\mathbb{C}[x]/(q')$, and the Chinese remainder theorem gives

$$\mathbb{C}[x]/(q') \simeq \bigoplus_{\tau} \mathbb{C}[x]/(x - \tau)^{m_\tau}.$$

For fixed τ , the images of $g_{\tau,k}$ vanish in every other local factor, while in the τ -factor, with $y = x - \tau$, they are

$$g_{\tau,k} = u_\tau(y) y^{m_\tau - k}, \quad u_\tau(0) \neq 0.$$

Hence $g_{\tau,1}, \dots, g_{\tau,m_\tau}$ form a basis of the τ -factor. Since $\sum_{\tau} m_\tau = n - 1$, all the displayed vectors form a basis of $\mathcal{V}$.

Since $q - \lambda_\tau$ vanishes at τ to order $m_\tau + 1$, the polynomial

$$h_{\tau,k}(x) = \frac{q(x) - \lambda_\tau}{(x - \tau)^k}$$

satisfies

$$qg_{\tau,k} = h_{\tau,k}q' + \lambda_{\tau}g_{\tau,k}.$$

Thus $C_*g_{\tau,k} = \lambda_{\tau}g_{\tau,k}$, which proves (ii) and the displayed decomposition of E_{λ} .

Write $m = m_{\tau}$, $y = x - \tau$, and

$$q(x) - \lambda_{\tau} = ay^{m+1} + O(y^{m+2}), \quad a \neq 0.$$

Then

$$g_{\tau,k} = a(m+1)y^{m-k} + O(y^{m-k+1})$$

and

$$-h'_{\tau,k} = -a(m+1-k)y^{m-k} + O(y^{m-k+1}).$$

In the ordered local basis, the error term involves only $g_{\tau,j}$ with $j < k$. Hence the compressed local block is triangular and its k th diagonal entry is $-(m+1-k)/(m+1)$.

If $\sigma \neq \tau$ and $\lambda_{\sigma} = \lambda_{\tau}$, then $h_{\tau,k} = (q - \lambda_{\tau})/(x - \tau)^k$ vanishes at σ to order $m_{\sigma} + 1$. Hence $h'_{\tau,k}$ vanishes modulo $(x - \sigma)^{m_{\sigma}}$. This excludes cross terms between distinct critical points over the same critical value and proves (iii). $\square$

Theorem 5.3 (Rational symmetry rigidity). *Let $P(z) \in M_{n-1}(\mathbb{C}(z))$ satisfy*

$$P' = [A(z), P]. \quad (5.11)$$

Then P is constant and commutes with both C and D .

Proof. First exclude finite poles. If P had a pole of order $m \geq 1$ at $z_0 \neq 0$, then P' would have pole order $m+1$, whereas the right side of (5.11) would have order at most m . Thus no such pole occurs.

At $z = 0$, suppose

$$P = z^{-m}P_{-m} + O(z^{-m+1}), \quad m \geq 1, \quad P_{-m} \neq 0.$$

The coefficient of z^{-m-1} is

$$-mP_{-m} = [D, P_{-m}].$$

By (3.8), every eigenvalue of $\text{ad } D : X \mapsto [D, X]$ lies strictly between -1 and 1 , so $-m$ cannot occur. Hence P has no finite pole and is a polynomial matrix.

Transpose (5.11) and put $Q = P^{\top}$. Then

$$Q' = [Q, C_* + D_*/z]. \quad (5.12)$$

Suppose

$$Q(z) = Q_d z^d + Q_{d-1} z^{d-1} + \cdots, \quad d \geq 1, \quad Q_d \neq 0.$$

Multiplying (5.12) by z and comparing the coefficients of z^{d+1} and z^d gives

$$[Q_d, C_*] = 0, \quad (5.13)$$

$$dQ_d = [Q_{d-1}, C_*] + [Q_d, D_*]. \quad (5.14)$$

For each eigenvalue λ of C_* , the spectral projection π_{λ} is a polynomial in C_* . Hence Q_d commutes with π_{λ} and is block diagonal. For a block $Q_{d,\lambda} = \pi_{\lambda}Q_d\pi_{\lambda} \neq 0$, compressing (5.14) to E_{λ} gives

$$dQ_{d,\lambda} = [Q_{d,\lambda}, \overline{D}_{\lambda}]. \quad (5.15)$$

After triangularizing $\overline{D}_\lambda$, the commutator on the right is triangular on matrix units, with diagonal entries equal to pairwise differences of eigenvalues of $\overline{D}_\lambda$. By (5.10), all these differences lie strictly between -1 and 1 . Equation (5.15) would make the positive integer d an eigenvalue, a contradiction. Thus $d = 0$.

Therefore P is constant. Substitution into (5.11) gives $[C, P] + z^{-1}[D, P] = 0$, hence $[C, P] = [D, P] = 0$. $\square$

Theorem 5.4 (Constant splitting). *Let $\mathcal{R} \subseteq \mathbb{C}(z)^{1 \times (n-1)}$ be stable under $\mathcal{T}$. Then there is a constant subspace $W \subseteq \mathbb{C}^{1 \times (n-1)}$, invariant under right multiplication by C and by D , such that*

$$\mathcal{R} = W \otimes_{\mathbb{C}} \mathbb{C}(z). \quad (5.16)$$

Proof. By Proposition 5.1, choose a stable complement of $\mathcal{R}$. Let right multiplication by $P(z)$ be the projector onto $\mathcal{R}$ along that complement. Since both summands are stable, the projector commutes with $\mathcal{T}$. For an arbitrary row ϕ , this identity is

$$\mathcal{T}(\phi P) = \mathcal{T}(\phi)P,$$

which is equivalent to

$$P' = [A(z), P].$$

By Theorem 5.3, P is constant and commutes with C, D . Taking W to be the constant row space of P gives (5.16); commutation with C, D gives the stated invariance. $\square$

6 From an algebraic value to an exponential polynomial

Let

$$\Lambda = \{0\} \cup \Lambda_\Delta \quad (6.1)$$

with repetitions removed, and put

$$\mathcal{E} = \text{span}_{\mathbb{C}(z)}\{e^{\lambda z} : \lambda \in \Lambda\}. \quad (6.2)$$

If $0 \notin \Lambda_\Delta$, its forcing vector is understood to be zero: $\mathbf{b}_0 = 0$.

The functions

$$U_0, \dots, U_{n-2}, \{e^{\lambda z} : \lambda \in \Lambda\}$$

form a vector of E -functions. Adjoining the exponential coordinates to (3.7) gives a homogeneous block system over $\overline{\mathbb{Q}}(z)$ with common denominator z . The coordinate $e^{0z} = 1$ homogenizes an algebraic value relation.

Define the rational relation space

$$\mathcal{R} = \left\{ \phi \in \mathbb{C}(z)^{1 \times (n-1)} : \phi \mathbf{U} \in \mathcal{E} \right\}. \quad (6.3)$$

It is stable under $\mathcal{T}$. Indeed, if

$$\mathbf{f}(z) = \frac{1}{nz} \sum_{\lambda \in \Lambda} \mathbf{b}_\lambda e^{\lambda z},$$

then (3.7) gives

$$(\phi \mathbf{U})' = \mathcal{T}(\phi) \mathbf{U} + \phi \mathbf{f}.$$

Both the left side and the final term lie in $\mathcal{E}$, so $\mathcal{T}(\phi) \in \mathcal{R}$.

Proposition 6.1 (An algebraic value forces an exponential polynomial). *Let $\xi \in \overline{\mathbb{Q}}^\times$. If $U_0(\xi) \in \overline{\mathbb{Q}}$, then*

$$U_0(z) = \sum_{\lambda \in \Lambda} p_\lambda(z) e^{\lambda z} \quad (6.4)$$

for polynomials $p_\lambda \in \mathbb{C}[z]$.

Proof. Let $a = U_0(\xi) \in \overline{\mathbb{Q}}$, order

$$\Lambda = \{0, \lambda_2, \dots, \lambda_s\},$$

and form

$$\mathbf{F}(z) = (U_0(z), \dots, U_{n-2}(z), 1, e^{\lambda_2 z}, \dots, e^{\lambda_s z})^\top.$$

Put

$$B_\Lambda = \frac{1}{n} \begin{pmatrix} \mathbf{b}_0 & \mathbf{b}_{\lambda_2} & \cdots & \mathbf{b}_{\lambda_s} \end{pmatrix}, \quad \Delta_\Lambda = \text{diag}(0, \lambda_2, \dots, \lambda_s).$$

Then

$$\mathbf{F}' = \begin{pmatrix} C + D/z & B_\Lambda/z \\ 0 & \Delta_\Lambda \end{pmatrix} \mathbf{F}. \quad (6.5)$$

The relation $U_0(\xi) - a = 0$ is homogeneous in the coordinates of $\mathbf{F}(\xi)$. More explicitly, let

$$\boldsymbol{\alpha} = (1, 0, \dots, 0; -a, 0, \dots, 0).$$

Then $\boldsymbol{\alpha} \mathbf{F}(\xi) = 0$. The matrix in (6.5) has common denominator z , and $\xi \neq 0$. By Theorem 2.2, there is a polynomial coefficient row

$$R(z) \in \overline{\mathbb{Q}}[z]^{1 \times ((n-1)+s)}$$

such that

$$R(z) \mathbf{F}(z) = 0, \quad R(\xi) = \boldsymbol{\alpha}.$$

Write

$$R(z) = (\phi(z); c_0(z), c_{\lambda_2}(z), \dots, c_{\lambda_s}(z)).$$

Then

$$\phi(z) \mathbf{U}(z) + c_0(z) + \sum_{j=2}^s c_{\lambda_j}(z) e^{\lambda_j z} = 0, \quad \phi(\xi) = \mathbf{e}_0 := (1, 0, \dots, 0). \quad (6.6)$$

Thus $\phi \in \mathcal{R}$.

By Theorem 5.4, $\mathcal{R} = W \otimes_{\mathbb{C}} \mathbb{C}(z)$ for a constant subspace W invariant under C, D . Equivalently, $\mathcal{R}$ is the image of a constant projector. Since $\phi(\xi) = \mathbf{e}_0$, we obtain $\mathbf{e}_0 \in W$ and hence

$$U_0 = \mathbf{e}_0 \mathbf{U} \in \mathcal{E}.$$

It remains to show that the rational exponential coefficients are polynomials. Choose a constant full-row-rank matrix S whose rows form a basis of W , and put $\mathbf{V} = S \mathbf{U}$. Invariance of W gives unique constant matrices C_W, D_W such that

$$SC = C_W S, \quad SD = D_W S.$$

With $\mathbf{b}_{W,\lambda} = S \mathbf{b}_\lambda$, the system (3.7) restricts to

$$\mathbf{V}' = \left(C_W + \frac{D_W}{z} \right) \mathbf{V} + \frac{1}{nz} \sum_{\lambda \in \Lambda} \mathbf{b}_{W,\lambda} e^{\lambda z}. \quad (6.7)$$

Every component of $\mathbf{V}$ lies in $\mathcal{E}$. By Lemma 2.3, there are unique rational vectors $\mathbf{r}_\lambda(z) \in \mathbb{C}(z)^{\dim W}$ such that

$$\mathbf{V}(z) = \sum_{\lambda \in \Lambda} \mathbf{r}_\lambda(z) e^{\lambda z}. \quad (6.8)$$

Equating exponential coefficients in (6.7) gives

$$\mathbf{r}'_\lambda = (C_W - \lambda I) \mathbf{r}_\lambda + \frac{D_W}{z} \mathbf{r}_\lambda + \frac{1}{nz} \mathbf{b}_{W,\lambda}. \quad (6.9)$$

A pole of $\mathbf{r}_\lambda$ at $z_0 \neq 0$ is impossible, because differentiation raises its order by one while the right side of (6.9) does not. At $z = 0$, suppose

$$\mathbf{r}_\lambda = z^{-m} v + O(z^{-m+1}), \quad m \geq 1, \quad v \neq 0.$$

The coefficient of z^{-m-1} gives

$$-mv = D_W v.$$

Every eigenvalue of D_W is an eigenvalue of D : if $u^\top D_W = \mu u^\top$, then $u^\top S \neq 0$ and

$$(u^\top S) D = u^\top D_W S = \mu (u^\top S).$$

By (3.8), D_W has no negative integral eigenvalue. Hence $\mathbf{r}_\lambda$ has no finite pole and belongs to $\mathbb{C}[z]^{\dim W}$.

Finally choose a constant row a_0 with $\mathbf{e}_0 = a_0 S$. Then

$$U_0 = \mathbf{e}_0 \mathbf{U} = a_0 \mathbf{V} = \sum_{\lambda \in \Lambda} (a_0 \mathbf{r}_\lambda(z)) e^{\lambda z},$$

which is (6.4) with $p_\lambda = a_0 \mathbf{r}_\lambda \in \mathbb{C}[z]$. □

Remark 6.2. Entirety alone does not imply polynomial exponential coefficients:

$$\frac{e^z - 1}{z} = \frac{1}{z} e^z - \frac{1}{z}$$

is entire although both displayed coefficients have a pole. This is the integral $\int_0^1 e^{zx} dx$ for a linear phase. For $n \geq 2$, the residue nonresonance in (3.8) is therefore load-bearing.

7 Constant-coefficient annihilation

Assume (6.4). Choose an integer N larger than the degrees of all nonzero polynomials p_λ ; with the convention $\deg 0 = -\infty$, choose

$$N > \max_{\lambda \in \Lambda} \deg p_\lambda.$$

Define the constant-coefficient differential operator

$$L = \prod_{\lambda \in \Lambda} (\partial_z - \lambda)^N. \quad (7.1)$$

Then $LU_0 = 0$. Since differentiation under the integral sign multiplies the integrand by $q(x)$,

$$0 = LU_0(z) = \int_{\Gamma} H(q(x)) e^{zq(x)} dx, \quad (7.2)$$

where

$$H(t) = \prod_{\lambda \in \Lambda} (t - \lambda)^N \quad (7.3)$$

is nonzero. Expanding at $z = 0$ gives

$$\int_{\Gamma} q(x)^m H(q(x)) dx = 0 \quad (m \geq 0). \quad (7.4)$$

The factor t^N in H arises from the constant coordinate used to homogenize the value relation; it need not have any geometric significance for q or Δ .

8 Cauchy-transform rigidity for phase moments

We now prove that the moment identities (7.4) force the underlying exponential period to vanish identically. The argument also gives an exact geometric criterion for functional relations.

Lemma 8.1 (Constellation representation). *Let $q \in \mathbb{C}[x]$ be nonconstant and let $\Delta = \sum_{\alpha \in S} d_{\alpha}[\alpha]$ be a finite complex zero-chain of degree zero. There exist:*

- (i) *a finite embedded tree T in the t -plane containing every finite critical value of q and every value $q(\alpha)$, $\alpha \in S$;*
- (ii) *a finite one-chain $\tilde{\Gamma}$, with complex coefficients, in the graph $q^{-1}(T)$ with $\partial \tilde{\Gamma} = \Delta$;*

such that, on every open edge e of T , the chain $\tilde{\Gamma}$ is a finite complex linear combination of analytic inverse branches of q over e .

Proof. Choose a regular center c and pairwise disjoint analytic arcs from c to the distinct finite critical values and to the distinct endpoint values $q(\alpha)$, omitting repeated arms. Their union is an embedded star and hence a finite tree T .

Over c , the preimage consists of the $n = \deg q$ sheets of the covering. The lifts of the critical-value arms join these sheets according to the branch monodromy generators. Since the polynomial covering is connected, its monodromy action is transitive, so the finite graph $q^{-1}(T)$ is connected. This is the standard polynomial constellation construction; compare [9, Section 2.1].

Fix a vertex x_* of $q^{-1}(T)$. For every $\alpha \in S$, choose an edge path P_{α} in $q^{-1}(T)$ from x_* to α , and put

$$\tilde{\Gamma} = \sum_{\alpha \in S} d_{\alpha} P_{\alpha}.$$

Because $\sum_{\alpha} d_{\alpha} = 0$, its boundary is

$$\partial \tilde{\Gamma} = \sum_{\alpha} d_{\alpha}[\alpha] - \left(\sum_{\alpha} d_{\alpha} \right) [x_*] = \Delta.$$

On the interior of every edge of T , the map q is unramified, so the stated inverse-branch decomposition follows. $\square$

Fix such a tree and graph chain. Orient every open edge e of T . If $n = \deg q$, let

$$x_{e,1}, \dots, x_{e,n} : e \longrightarrow \mathbb{C}, \quad q(x_{e,i}(t)) = t,$$

be the analytic inverse branches. Let $f_{e,i} \in \mathbb{C}$ be the coefficient with which $\tilde{\Gamma}$ traverses the lift $x_{e,i}(e)$, relative to the chosen orientation, and define the edge density

$$\psi_e(t) = \sum_{i=1}^n f_{e,i} x'_{e,i}(t) = \sum_{i=1}^n \frac{f_{e,i}}{q'(x_{e,i}(t))}. \quad (8.1)$$

These densities are analytic on the open edges. Their endpoint singularities are integrable. Indeed, if $t = c$ is an endpoint of e , a branch tends to a point $\zeta \in q^{-1}(c)$ of ramification index r , and

$$q(x) - c = (x - \zeta)^r u(x), \quad u(\zeta) \neq 0.$$

Along the edge,

$$x_{e,i}(t) - \zeta = O((t - c)^{1/r}), \quad x'_{e,i}(t) = O(|t - c|^{1/r-1}),$$

and $1/r - 1 > -1$.

Theorem 8.2 (Phase-moment rigidity and edge-density criterion). *Let $q \in \mathbb{C}[x]$ be nonconstant, let Δ be a finite complex zero-chain of degree zero, let Γ be any one-chain with boundary Δ , and let $0 \neq H \in \mathbb{C}[t]$. Suppose*

$$\int_{\Gamma} q(x)^m H(q(x)) dx = 0 \quad (m \geq 0). \quad (8.2)$$

Then every edge density (8.1) of the constellation representation in Lemma 8.1 vanishes identically. Consequently,

$$I_{q,\Delta}(z) \equiv 0. \quad (8.3)$$

In particular, for a fixed constellation representation,

$$I_{q,\Delta}(z) \equiv 0 \iff \psi_e \equiv 0 \text{ for every open edge } e \subset T. \quad (8.4)$$

Proof. Every differential $q(x)^m H(q(x)) dx$ is polynomial and hence has a polynomial primitive. Its integral depends only on the boundary. Therefore (8.2) remains valid after replacing Γ by the graph chain $\tilde{\Gamma}$ from Lemma 8.1.

For $s \in \mathbb{C} \setminus T$, define the Cauchy transform

$$\mathcal{C}_H(s) = \int_{\tilde{\Gamma}} \frac{H(q(x))}{s - q(x)} dx. \quad (8.5)$$

The path is compact and piecewise analytic, and the denominator is bounded away from zero locally uniformly on $\mathbb{C} \setminus T$. Thus $\mathcal{C}_H$ is holomorphic there. Put

$$M = \max_{x \in \tilde{\Gamma}} |q(x)|.$$

For $|s| > M$, the geometric series

$$\frac{1}{s - q(x)} = \sum_{m \geq 0} \frac{q(x)^m}{s^{m+1}}$$

converges uniformly on $\tilde{\Gamma}$. Using (8.2), we obtain

$$\mathcal{C}_H(s) = \sum_{m \geq 0} \frac{1}{s^{m+1}} \int_{\tilde{\Gamma}} q(x)^m H(q(x)) dx = 0 \quad (|s| > M).$$

The complement of a finite embedded tree in the plane is connected. Hence the identity theorem gives

$$\mathcal{C}_H \equiv 0 \quad \text{on } \mathbb{C} \setminus T. \quad (8.6)$$

Changing variables $t = q(x)$ separately on every lifted edge gives the absolutely convergent representation

$$\mathcal{C}_H(s) = \sum_e \int_e \frac{H(t)\psi_e(t)}{s-t} dt, \quad s \in \mathbb{C} \setminus T. \quad (8.7)$$

Fix an open edge e and an interior point $t_0 \in e$ with $H(t_0) \neq 0$. Choose a compact analytic subarc $\eta \Subset e$ containing t_0 in its interior and a disk D about t_0 such that

$$\overline{D} \cap (T \setminus \eta) = \emptyset.$$

After shortening η , all inverse branches occurring in ψ_e extend holomorphically to a neighborhood of η . Thus

$$\rho(t) = H(t)\psi_e(t)$$

is holomorphic there. The contribution to (8.7) from $T \setminus \eta$ extends holomorphically across D , so the entire lateral jump of $\mathcal{C}_H$ across η is the jump of

$$J(s) = \int_\eta \frac{\rho(t)}{s-t} dt.$$

Let a, b be the initial and terminal endpoints of the oriented arc η . For s near t_0 , define

$$K(t, s) = \begin{cases} \frac{\rho(t) - \rho(s)}{t - s}, & t \neq s, \\ \rho'(s), & t = s. \end{cases}$$

Then K is jointly holomorphic near $\eta \times \eta$, and

$$J(s) = \rho(s) \int_\eta \frac{dt}{s-t} - \int_\eta K(t, s) dt. \quad (8.8)$$

The second term is holomorphic across η . On either side of the arc, the first scalar integral is a branch of

$$\int_\eta \frac{dt}{s-t} = \log(s-a) - \log(s-b).$$

Its two lateral determinations differ by $2\pi i \varepsilon_e$, where $\varepsilon_e \in \{\pm 1\}$ depends only on the orientation and the designation of the two sides. Hence

$$(\mathcal{C}_H)_+(t_0) - (\mathcal{C}_H)_-(t_0) = 2\pi i \varepsilon_e H(t_0)\psi_e(t_0). \quad (8.9)$$

By (8.6), both lateral values vanish. Since $H(t_0) \neq 0$, we conclude that $\psi_e(t_0) = 0$. The point t_0 was arbitrary outside the finite zero set of H , and ψ_e is analytic on the open edge. Therefore

$$\psi_e \equiv 0$$

for every open edge e .

Finally, changing variables edge by edge in the exponential integral gives

$$I_{q,\Delta}(z) = \int_{\tilde{\Gamma}} e^{zq(x)} dx = \sum_e \int_e e^{zt} \psi_e(t) dt = 0.$$

This proves (8.3). If $I_{q,\Delta} \equiv 0$, its Taylor coefficients at $z = 0$ give (8.2) with $H = 1$, so the forward implication in (8.4) follows from the theorem. The reverse implication is the final displayed edge decomposition. $\square$

9 Proofs of the main theorems

Proof of Theorem 1.2. The implication $I_{q,\Delta} \equiv 0 \Rightarrow I_{q,\Delta}(\xi) \in \overline{\mathbb{Q}}$ is immediate. We prove the converse.

Suppose first that $q(x) = ax + b$ with $a \neq 0$. Since $e^{zq(x)}/(az)$ is a primitive for $z \neq 0$, one has

$$I_{q,\Delta}(z) = \frac{1}{az} \sum_{\alpha \in S} d_\alpha e^{zq(\alpha)}. \quad (9.1)$$

Assume that $I_{q,\Delta}(\xi) = \gamma \in \overline{\mathbb{Q}}$. Then

$$\sum_{\alpha \in S} d_\alpha e^{\xi q(\alpha)} - a\xi\gamma e^0 = 0. \quad (9.2)$$

The map q is injective, so the exponents $\xi q(\alpha)$ are pairwise distinct. After combining the coefficient of e^0 if one of them is zero, Theorem 2.4 shows that $d_\alpha = 0$ whenever $q(\alpha) \neq 0$. If there is an α_0 with $q(\alpha_0) = 0$, the only remaining coefficient relation is $d_{\alpha_0} = a\xi\gamma$; but the degree-zero condition $\sum_\alpha d_\alpha = 0$ gives $d_{\alpha_0} = 0$, hence $\gamma = 0$. If no such α_0 exists, (9.2) directly gives $\gamma = 0$. Thus every d_α is zero, so $I_{q,\Delta} \equiv 0$.

Now suppose $\deg q = n \geq 2$. In the notation of (3.3), one has $U_0 = I_{q,\Delta}$. If $U_0(\xi)$ is algebraic, then Proposition 6.1 gives the exponential-polynomial representation (6.4). The annihilation argument of Section 7 produces a nonzero polynomial H satisfying the phase-moment identities (7.4). Applying Theorem 8.2 yields $U_0 \equiv 0$. This proves the converse implication and the theorem. $\square$

Proof of Corollary 1.3. Take $\Delta = [B] - [A]$. Since

$$I_{q,\Delta}(0) = \int_A^B 1 \, dx = B - A \neq 0,$$

the entire function $I_{q,\Delta}$ is not identically zero. Apply Theorem 1.2. $\square$

Proof of Theorem 1.4. Let

$$\Delta = \sum_{j=1}^r a_j \Delta_j.$$

Then $I_{q,\Delta} = \sum_{j=1}^r a_j I_{q,\Delta_j}$. If the left side of (1.5) vanishes, then

$$I_{q,\Delta}(\xi) = -a_0 \in \overline{\mathbb{Q}}.$$

By Theorem 1.2, $I_{q,\Delta} \equiv 0$ and its value at ξ is zero. Therefore $a_0 = 0$. The converse is immediate. The dimension identity (1.6) is the corresponding equality of relation spaces. $\square$

Lemma 9.1 (Separated regular fibers give functional independence). *Let $a_0, \dots, a_r \in \mathbb{C}$ be such that $q(a_0), \dots, q(a_r)$ are pairwise distinct regular values of q . If*

$$\Delta = \sum_{i=0}^r d_i [a_i], \quad \sum_{i=0}^r d_i = 0,$$

and $I_{q,\Delta} \equiv 0$, then $d_0 = \dots = d_r = 0$.

Proof. Choose the star in Lemma 8.1 so that every $t_i = q(a_i)$ is the terminal endpoint of its own arm. Because t_i is a regular value, every point of $q^{-1}(t_i)$ is unramified, and every lift of the terminal arm has exactly one incident edge at its terminal vertex. Orient the terminal arm toward t_i and use a graph chain $\tilde{\Gamma}$ with boundary Δ .

Let $x_{i,1}, \dots, x_{i,n}$ be the inverse branches over the open terminal arm, numbered so that $x_{i,1}(t) \rightarrow a_i$ as $t \rightarrow t_i$. If $f_{i,k}$ is the coefficient of the corresponding lifted edge in $\tilde{\Gamma}$, then the boundary coefficient at its terminal vertex is exactly $f_{i,k}$. Since the only point of the support of Δ in the fiber over t_i is a_i , one has

$$f_{i,1} = d_i, \quad f_{i,k} = 0 \quad (k \geq 2).$$

By the edge-density criterion (8.4), the density on this arm vanishes identically. Hence

$$0 = \psi_i(t) = \frac{d_i}{q'(x_{i,1}(t))}.$$

The denominator is nonzero, so $d_i = 0$. This holds for every i . □

Proof of Corollary 1.5. Put

$$I_j(z) = \int_{a_0}^{a_j} e^{zq(x)} dx, \quad 1 \leq j \leq r.$$

If $\sum_{j=1}^r c_j I_j(z) \equiv 0$, then the corresponding boundary zero-chain is

$$\sum_{j=1}^r c_j ([a_j] - [a_0]) = \sum_{j=1}^r c_j [a_j] - \left(\sum_{j=1}^r c_j \right) [a_0].$$

By Lemma 9.1, all its coefficients vanish, and therefore $c_1 = \dots = c_r = 0$. Thus $I_1, \dots, I_r$ are linearly independent over $\overline{\mathbb{Q}}$ as entire functions. Apply Theorem 1.4. □

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